## FACULTY OF ENGINEERING, SCIENCES AND TECHNOLOGY

| (Problem Based Learning Assignment) | | |
|---|---|---|
| **CLOs** | **Blooms Taxonomy** | **GAs** |
| CLO1<br>CLO2 | C3<br>P2 | GA-2<br>GA-5 |

| **Assigned Week** | 7th Week | **Assessment Week** | 13th Week |
|---|---|---|---|

| **WP** | **Characteristics** | **Description** |
|---|---|---|
| WP1<br>WP3<br>WP4 | Range of conflicting requirements<br>Depth of knowledge required<br>Familiarity of issues | Solving a real-world problem by predicting diabetes risk using AI techniques, integrating machine learning models for risk prediction based on patient health data.<br><br>Requires understanding of AI modeling, data preprocessing, and performance evaluation to train and compare multiple machine learning models effectively.<br><br>Involves infrequently encountered issues, such as dealing with missing data, selecting relevant features, tuning models for optimization, and ensuring real-time prediction accuracy. |

## 1. Problem Statement

In this **challenge**, the **goal** is **design** and **implement** a working AI-based solution to a real-world (or realistically simplified) problem, using techniques practiced in the lab manual, while being free to research and add extra techniques that make the solution stronger.

## 2. Objective

- Identify a real-world problem that can be solved using one or more AI techniques covered in the course.
- Formulate the problem the way it was practiced in labs (states, agents, rules, fuzzy variables, or training data, depending on the chosen technique).
- Implement the solution in Python, reusing and extending the algorithms built during the lab sessions.

## 4. Topic Options

- Option A — Intelligent Pathfinding & Route Optimization Agent
- Option B — Fuzzy Logic-Based Decision / Control System
- Option C — Game-Playing Agent with Adversarial Search
- Option D — Predictive / Classification System using Machine Learning

## 3. Features and Functionalities

- Clear problem formulation stating states/inputs, actions/rules/features, and goal/output.
- A working Python implementation of the chosen AI technique(s).
- A performance or correctness evaluation appropriate to the technique (e.g. nodes expanded and path cost for search; accuracy/precision/recall for ML; rule coverage and output sensitivity for fuzzy systems).
- A simple way to interact with the system (console input, or a small GUI).

## 4. Student Guidelines

1. Understand dataset and preprocess data.
2. Implement machine learning algorithms.
3. Optimize model performance using tuning techniques.
4. Write pseudocode before coding the model.

## 5. Pseudocode for AI Model

**General problem-solving pseudocode:**

```
DEFINE problem (states/inputs, actions/rules, goal/output)
COLLECT or CONSTRUCT the data/graph/rule-base needed
FOR each candidate technique/algorithm chosen:
   RUN the algorithm on the problem
   RECORD performance metric(s)
```

SELECT best-performing / most suitable approach
BUILD a simple interface to accept new input and produce output
EVALUATE and DISPLAY results with justification

**6. Flowchart**

Students should design a flowchart showing the full pipeline of their chosen option.

For example: problem/data input → preprocessing or graph construction → algorithm execution (search / fuzzy rule evaluation / model training) → evaluation → output/decision.

The flowchart should be included in the submitted report.

**7. Submission Checklist**

- Source Code
- Presentation